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Roll No

CE-302 (GS)

B.E. III Semester Examination, June 2020

Grading System (GS)

Transportation Bridges and Tunnels

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. How other engineering surveys are carried out for railway alignment?
2. State all types of Indian loading standards for railway bridges and highway bridge with suitable sketch.
3. Calculate curve lead, radius, switch lead and lead to set out a 1 in $8\frac{1}{2}$ turnout taking off from a straight B.G. track with its curve starting from the toe of switch and passes through the Theoretical nose of crossing head divergence $d=11.4\text{cm}$.
4. Describe different types of foundation in brief. Describe with neat sketch all the working components of Cofferdam.
5. Describe following in details. (any two)
 - a) Super elevation and cant deficiency
 - b) Negative super elevation
 - c) Permissible speed on a horizontal curve
6. Describe following in details (any one)
 - a) Requirements of good ballast
 - b) Tractive resistance and its expressions
7. Compare the different types of rails with the help of neat sketches.

OR

Explain the requirements of a good sleeper. What is sleeper density?

8. Describe in brief: (any two)
 - a) Economical span of a bridge.
 - b) Scour depth and linear water way.
 - c) Choice of type of foundation of bridge.
 - d) Type of bearings and their functions.
